R823-HG-T005

Isomorfia e cambio de base: §4, parte 1b

Status: provisional construction test. This is a source-keyed translation experiment, not a native-reviewed standard, not an intelligibility result, and not a completed translation of the R823 volume.

Source boundary. German cumulative authority, lines 21148–21202. Authority SHA-256 begins `EE8955BCF7A2639`; exact slice SHA-256 begins `90FBAF614F9DDF01`. Every translated source segment is aligned in the accompanying clause map. Lines 21204–21208 are enumerate scaffolding; the next semantic source unit begins at line 21209 with theorem 2.

§ 4 (continuation): Isomorfia e cambio de base

Le isomorfia inter $\overline{\mathfrak{D}}$ e le representation es explicate per le formulas sequente. Si al transformation linear generate per $c \in \mathfrak{o}$ corresponde le matrice C, dunque

$$c(x_1, \dots, x_n) = (x_1, \dots, x_n)C,$$

e si, in maniera analoge, al elemento $d \in \mathfrak{o}$ corresponde le matrice $D \in \mathsf{T}_n$,

$$d(x_1, \dots, x_n) = (x_1, \dots, x_n)D,$$

in tal caso evidentemente

$$\begin{aligned}(c+d)(x_1, \dots, x_n) &= (cx_1 + dx_1, \dots, cx_n + dx_n)\\ &= (x_1, \dots, x_n)(C + D).\end{aligned}$$

Et etiam

$$\begin{aligned}cd(x_1, \dots, x_n) &= c(dx_1, \dots, dx_n) = c((x_1, \dots, x_n)D)\\ &= (cx_1, \dots, cx_n)D = (x_1, \dots, x_n)CD.\end{aligned}$$

Assi, le transformationes linear de $\mathfrak{M}$ a se mesme generate per $\mathfrak{o}$, realisate mediante un base determinate x_ν de $\mathfrak{M}$, dona un representation de grado n de $\mathfrak{o}$ in T.

Si $(y_1, \ldots, y_n)$ es un altere base de $\mathfrak{M}$, illo deriva de $(x_1, \ldots, x_n)$ per multiplication con un matrice regular P:

$$(y_1, \ldots, y_n) = (x_1, \ldots, x_n)P.$$

Si un transformation linear de $\mathfrak{M}$ es realisate per le matrice C relative al base x_ν, relative al base y_ν illo es realisate per $P^{-1}CP$. Nam, si

$$c(x_1, \ldots, x_n) = (x_1, \ldots, x_n)C,$$

in tal caso

$$\begin{aligned} c(y_1, \ldots, y_n) &= c(x_1, \ldots, x_n)P = ((x_1, \ldots, x_n)C)P \\ &= (x_1, \ldots, x_n)CP = ((y_1, \ldots, y_n)P^{-1})CP \\ &= (y_1, \ldots, y_n)P^{-1}CP. \end{aligned}$$

De isto se conclude directmente: *Le modul $\mathfrak{M}$ dona tote le representationes de un classe de representation, si tote le bases de $\mathfrak{M}$ es usate pro realisar le transformationes linear de $\mathfrak{M}$ inducite per $\mathfrak{o}$.*

Nota editorial de senso

Le tupla ordinate $(x_1, \ldots, x_n)$ es tractate como un vector-riga, e le matrices acta al dextra. Le pares $c \mapsto C$ e $d \mapsto D$ resta fixe. Dunque $(c + d) \mapsto C + D$ e $cd \mapsto CD$; le ordine CD non es invertite per un convention moderne de composition.

Le cambio de base ha le direction $(y_1, \ldots, y_n) = (x_1, \ldots, x_n)P$. Per iste direction, le matrice del mesme transformation in le base y_ν es $P^{-1}CP$, non PCP^{-1}. Le forma historic *matrice regular* es limitate hic al senso de matrice invertibile.

Le conclusion dice que variar tote le bases de $\mathfrak{M}$ dona tote le representationes *de un classe de representation.* Isto non affirma que tote le matrices de T_n es obtenite, ni que $\mathfrak{M}$ produce tote le classes de representation. Solmente le matrices del transformationes inducite per $\mathfrak{o}$ e exprimite in le bases de $\mathfrak{M}$ es in scope.

Limites del tranche

Le grammatica e le vocabulario es candidatos de construction. Nulle affirmation de comprehension human, de validation native, o de intelligibilitate marginal es facite. Il ha zero observationes human e nulle resultato pilot. Le proxime unitate semantic comencia al linea 21209 con theorem 2; le lineas 21204–21208 es solmente le structura LaTeX de su enumeration.